Medical Question Answering using Llama 3 + RAG

A Retrieval-Augmented Generation System for Grounded, Trustworthy Clinical Decision Support

Project Team

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Project Overview

This project explores the integration of Meta's **Llama 3** large language model with a **Retrieval-Augmented Generation (RAG)** pipeline to enable safe, accurate, and source-grounded medical question answering.

Evaluated on the **MedQA (USMLE)** benchmark using a corpus of 18 medical textbooks, this system directly addresses the hallucination and knowledge-staleness problems endemic to standalone LLMs in clinical settings.

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MOTIVATION

Why Medical QA Demands a Better Approach

Large language models have demonstrated remarkable fluency — but fluency without factual grounding is dangerous in clinical contexts. Four interconnected problems drive this project.

🚨 LLMs Hallucinate

Models like Llama 3 are powerful but routinely generate plausible-sounding yet factually false medical information. In clinical contexts, this is not a minor inconvenience — it is a patient safety risk that makes unaugmented LLMs unsuitable for medical use.

📚 Fine-Tuned Models Lack Grounding

ChatDoctor and similar systems improve conversational quality by fine-tuning on patient-doctor dialogues, but they rely on **static, memorized knowledge** without real-time verification. Knowledge becomes stale, and the model cannot cite where its answer came from.

✅ RAG Solves the Grounding Problem

Retrieval-Augmented Generation retrieves verified medical documents *before* generation, ensuring answers are conditioned on authoritative, up-to-date sources. This fundamentally shifts the system from memorization to evidence-based reasoning.

🚀 Career & Research Relevance

Medical AI and health-tech are among the fastest-growing fields in applied NLP. RAG-based clinical systems represent the frontier of safe, trustworthy decision support — making this project directly relevant to both academic and industry career trajectories.

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CORE CONCEPT

The Llama 3 + RAG Pipeline

Llama 3 generates natural language answers to medical questions, while a RAG pipeline retrieves relevant medical text from a vector database *before* generation. This grounds responses in verifiable sources instead of relying purely on memorized parametric knowledge.



Key Design Principle

Every answer generated by the system is **conditioned on retrieved context** — not on the model's internal weights alone. This architectural choice is the primary mechanism for hallucination reduction.

Why This Architecture?

Unlike fine-tuned models that bake knowledge into parameters at training time, a RAG system can be **updated by swapping the retrieval corpus** — no retraining required. This makes it inherently more maintainable and adaptable to new medical evidence.

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DATASET

Benchmark & Knowledge Base Selection

The choice of datasets directly determines the credibility of evaluation and the quality of retrieved context. Both selections are grounded in established literature conventions.

QA Benchmark: MedQA (USMLE)

USMLE-style multiple-choice medical questions representing the standard benchmark for evaluating medical LLMs. Questions require clinical reasoning across a broad range of medical topics, mirroring real licensing exam difficulty.

- Multi-choice format aligned with real clinical exam standards
- Widely used — enables direct, fair comparison with published results
- De facto standard for medical QA evaluation in the NLP literature

RAG Corpus: Medical Textbooks + PubMed

The retrieval knowledge base is constructed from MedQA's accompanying **18 medical textbooks**, with optional extension via **PubMed abstracts** for broader coverage of recent biomedical research.

- 18 curated textbooks provide structured, authoritative clinical knowledge
- PubMed abstracts extend coverage to recent research findings
- Corpus directly matches benchmark domain — retrieval is contextually relevant by design

Why these datasets: Both are widely benchmarked in the literature, enabling fair comparison with existing systems. MedQA is the de facto standard, ensuring our results are interpretable and reproducible within the community.

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LITERATURE SURVEY · PART 1 OF 4

RAG Architectures for Medical QA

The following five works represent key developments in retrieval-augmented and hybrid approaches to medical NLP, spanning evaluation frameworks, filtering strategies, and iterative retrieval.

MedRGB — Ngo Trung et al., 2024

Approach: Evaluation framework testing RAG systems under realistic failure scenarios — noisy documents, conflicting evidence, and multi-hop reasoning.
Advantage: Tests robustness beyond standard retrieve-then-answer pipelines.
Limitation: Framework-focused only; proposes no new model architecture.

RAG² / Rationale-Guided RAG — Sohn et al., NAACL 2025

Approach: Filtering model selects informative snippets; LLM-generated rationales serve as refined retrieval queries, reducing noise in retrieved context.
Advantage: Directly tackles irrelevant and noisy retrieval — a major failure mode in standard RAG.
Limitation: Adds pipeline complexity and inference overhead.

Iterative Follow-up Questions — Xiong et al., PSB 2025

Approach: When initial retrieval is insufficient, the system generates follow-up queries to iteratively gather additional relevant context.
Advantage: Handles complex, multi-step clinical questions that require chained reasoning.
Limitation: Multiple retrieval rounds increase system latency significantly.

MKRAG — Shi et al., AMIA 2024

Approach: Retrieval-augmented generation applied to Vicuna-7B without fine-tuning; improved accuracy from **44.46% → 48.54%** on medical benchmarks.
Advantage: Demonstrates RAG benefit without any fine-tuning overhead.
Limitation: Older, smaller model; modest gain of ~4 percentage points.

QLoRA + RAG — 2025

Approach: Combines quantized low-rank adaptation (QLoRA) fine-tuning with a RAG retrieval layer for resource-constrained deployment.
Advantage: Resource-efficient — viable for smaller institutions without high-end GPU clusters.
Limitation: Trade-off between quantization efficiency and full model capability.

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LITERATURE SURVEY · PART 2 OF 4

Fine-Tuning Approaches for Medical LLMs

Five works examining how fine-tuning strategies — from chain-of-thought instruction tuning to domain-specific dialogue training — shape medical LLM performance, along with their key limitations.

Fine-Tuning LLaMA-3.2-3B for Medical CoT — 2025

Approach: Chain-of-thought fine-tuning applied to the smaller 3B parameter variant of the LLaMA-3.2 family, optimizing reasoning traces for medical questions.
Advantage: Substantially lower compute cost relative to 8B+ models.
Limitation: Smaller model trades off reasoning depth on complex multi-step clinical questions.

ChatDoctor — Li et al., Cureus 2023

Approach: LLaMA fine-tuned on 100k patient-doctor dialogues with a secondary self-directed retrieval module drawing from Wikipedia and medical databases.
Advantage: Outperforms ChatGPT on precision, recall, and F1 for medical dialogue tasks.
Limitation: Retrieval is a secondary add-on, not core design; core knowledge remains static and memorized.

Fine-Tuning Llama 3 for Physician Letters — Frontiers in AI, 2024

Approach: Local fine-tuning of Llama 3 for privacy-preserving clinical letter generation without cloud data transmission.
Advantage: Strong privacy compliance — runs entirely on-premise.
Limitation: Narrow task scope (letter generation only); not designed for general-purpose medical QA.

Instruction Tuning in Biomedical NLP — 2024

Approach: Ablation study systematically examining instruction tuning strategies across biomedical NLP tasks to identify which components drive performance gains.
Advantage: Provides actionable insights into effective tuning approaches for the biomedical domain.
Limitation: Analysis-focused; no deployable system is produced.

Medical Transcription in Low-Resource Languages — 2025

Approach: LLaMA 3.1-8B fine-tuned on Finnish clinical dialogues, achieving **BERTScore 0.82** despite low BLEU scores — demonstrating semantic adequacy over surface-level match.
Advantage: High semantic similarity on a language the model was not natively pre-trained on.
Limitation: Extremely small dataset (7 samples); generalizability is severely limited.

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LITERATURE SURVEY · PART 3 OF 4

Benchmarks, Datasets & RAG Evaluation Frameworks

Five works establishing the evaluation landscape for medical QA — from foundational datasets to systematic RAG component benchmarking — that directly inform our experimental design.

MedExQA — 2024

Approach: Benchmark requiring models to produce multiple explanation styles alongside answers, moving beyond single-token accuracy to explainability assessment.
Advantage: Tests explainability — critical for clinical trust and regulatory acceptance.
Limitation: Subjective explanation quality is difficult to evaluate automatically at scale.

PubMedQA — Jin et al., EMNLP-IJCNLP 2019

Approach: Biomedical QA from PubMed abstracts: **1k expert-annotated**, 61.2k unlabeled, 211.3k synthetic instances. Requires reasoning over quantitative research findings.
Advantage: First QA dataset demanding reasoning over quantitative biomedical data; pairs naturally with RAG retrieval of PubMed content.
Limitation: Best model at **68.1%** vs. **78% human performance**; yes/no/maybe format is simplistic for complex clinical scenarios.

Benchmarking RAG for Medicine — Xiong et al., ACL Findings 2024

Approach: Systematic benchmark of RAG pipeline components — retrievers, corpora, and model sizes — to identify which configurations drive medical QA performance.
Advantage: Directly relevant for pipeline design; provides evidence-based component selection guidance.
Limitation: Benchmark-only study; no new architecture is proposed.

LLM Benchmarks in Medical Tasks — Survey, 2024

Approach: Comprehensive survey of medical LLM benchmarks — MedQA, MedMCQA, PubMedQA, and others — providing a unified reference for the evaluation landscape.
Advantage: Single reference for benchmark comparison and selection across projects.
Limitation: Literature review only; no original experiments conducted.

MedExpQA — ScienceDirect, 2024

Approach: Multilingual medical QA benchmarking, evaluating LLM performance across non-English languages to expose performance disparities.
Advantage: Highlights critical non-English performance gaps in medical AI — important for global health equity.
Limitation: Multilingual scope is outside the focus of this project.

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LITERATURE SURVEY · PART 4 OF 4

Hallucination Detection & Clinical Safety

Five works addressing the hallucination problem in medical LLMs — a research thread directly motivating this project's use of RAG as a structural intervention against factual drift.

MedHallBench — 2024

Approach: Purpose-built benchmark for measuring hallucination rates in medical LLMs across diverse clinical question types.
Advantage: First benchmark specifically designed for medical hallucination measurement — fills a critical evaluation gap.
Limitation: Detection-only framework; proposes no reduction or mitigation strategies.

MedHal — 2024

Approach: Hallucination detection dataset constructed on top of PubMedQA, pairing correct answers with synthetically generated hallucinated alternatives.
Advantage: Integrates seamlessly with PubMedQA-based experimental setups.
Limitation: Synthetically generated hallucinations may not accurately reflect real-world LLM failure patterns.

Risk-Sensitive Hallucination Score (RSHS) — 2024

Approach: RSHS framework detecting high-risk language patterns — treatment directives, urgency cues, and over-confident clinical recommendations — using keyword heuristics.
Advantage: Exposes dangerous "high-risk, low-grounding" failures that standard accuracy metrics miss entirely.
Limitation: Keyword heuristics may miss nuanced risks; synthetic prompts limit real-world clinical applicability.

Rethinking RAG for Medicine — Large-Scale Expert Evaluation, 2025

Approach: Systematic expert clinical evaluation of RAG systems, replacing automated metrics with physician-level quality assessment.
Advantage: Human evaluation is far more clinically credible than automated metrics like BLEU or accuracy.
Limitation: Expert evaluation is expensive, slow, and difficult to scale for continuous benchmarking.

Hallucinations & Key Information Extraction in Medical Texts — 2025

Approach: Assesses hallucination rates across multiple open-source LLMs including base and fine-tuned variants, evaluating key clinical information extraction fidelity.
Advantage: Directly comparable to the Llama 3 experimental setup used in this project.
Limitation: Evaluates base and fine-tuned models only — does not assess RAG-augmented configurations.

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RESEARCH GAP & OBJECTIVE

What the Literature Leaves Unresolved

Three Critical Gaps in Existing Work

1 Static Knowledge in Fine-Tuned Models

Fine-tuned medical LLMs (ChatDoctor, CoT-tuned variants) embed knowledge into model weights at training time. This knowledge becomes stale and cannot be updated without costly retraining — and remains prone to hallucination with no verification mechanism.

2 Outdated Base Models in RAG Systems

Most existing RAG-medicine systems — including MKRAG (Vicuna-7B) and earlier GPT-3.5-based pipelines — use older or smaller LLMs as the generator. Newer, stronger open-source models like **Llama 3** have not been systematically evaluated in a full RAG configuration on MedQA.

3 Missing Source Attribution

Few systems combine factual grounding with explicit source attribution. For clinical trust and regulatory acceptability, transparency about *where* an answer comes from is as important as whether the answer is correct.

Project Objective

> To develop a Retrieval-Augmented Generation system using Llama 3 that improves factual accuracy and reduces hallucination in medical question answering by grounding responses in verified medical sources.

How the Objective Addresses the Gaps

- **Gap 1 → RAG grounding:** Every answer is conditioned on retrieved text — not static weights
- **Gap 2 → Llama 3 as generator:** Modern, strong open-source LLM applied in full RAG configuration
- **Gap 3 → Source attribution:** Evaluation explicitly measures retrieval relevance and grounding transparency

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PROPOSED METHODOLOGY

How This Project Improves on Existing Work

A direct comparison of this project's design choices against the limitations identified across 20+ surveyed papers demonstrates clear, targeted improvements along three dimensions.

Existing Work	Limitation Identified	Our Project's Improvement
ChatDoctor / Fine-tuned LLaMA variants	No retrieval; static fine-tuned knowledge baked into weights; prone to hallucination with no verification mechanism	**RAG grounds every answer** in retrieved medical text — knowledge is external, verifiable, and updateable without retraining
MKRAG, older RAG systems (Vicuna-7B, GPT-3.5-based pipelines)	Use smaller or older LLMs as the generator; capability ceiling limits answer quality on complex clinical questions	**Uses Llama 3** — a newer, stronger open-source generator that brings state-of-the-art reasoning to the RAG pipeline
Most medical LLM evaluation papers	Evaluate only on accuracy metrics (% correct); no assessment of retrieval quality or source transparency	**Also evaluates retrieval relevance and source grounding** — enabling transparency and clinical trust beyond raw accuracy

20+

Papers Surveyed

Across RAG, fine-tuning, hallucination, and benchmarking literature

3

Key Gaps Addressed

Static knowledge, outdated generators, and missing source attribution

18

Textbooks in Corpus

MedQA's curated medical textbooks form the RAG retrieval knowledge base

Llama 3

Generator Model

Newest open-source LLM applied in a full RAG configuration on MedQA

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